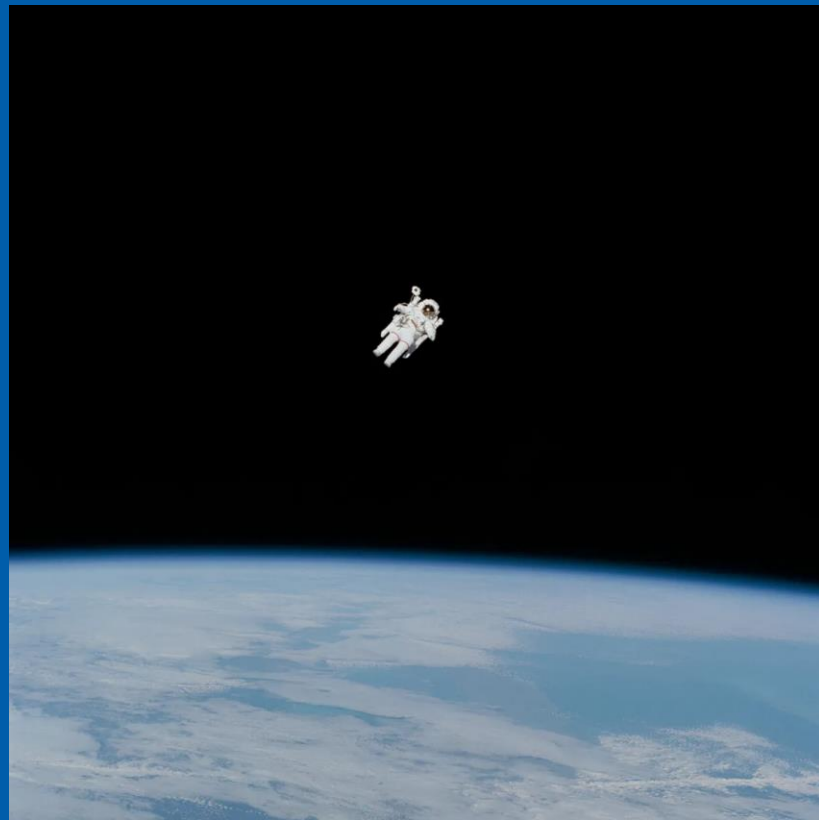
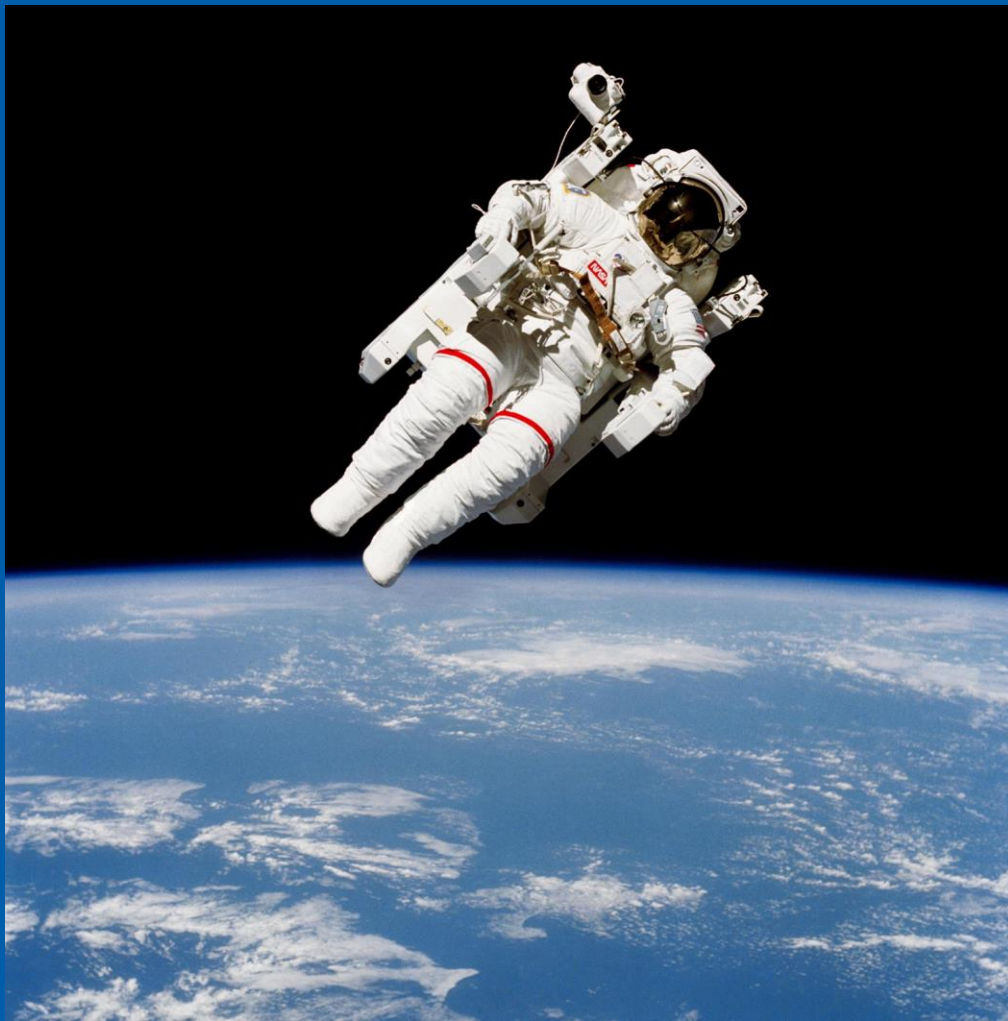


Astronomy 102 - Lab 01

- Scales of the Universe



Astronaut Bruce McCandless II flew the Manned Maneuvering Unit out of the space shuttle Challenger's payload bay for the first time on February 7, 1984.

Course Overview

What do you think of when you hear the word astronomy?

Why did you choose this class? If it fulfills a requirement, why this course over other options?

Course Overview

What do you hope to learn throughout this semester?

Can you recall hearing or seeing astronomy in the news?

How do you see the relationship between math and science/astronomy?

The latest in Astronomy

Honouring the legacy of Dr. Gladys West:

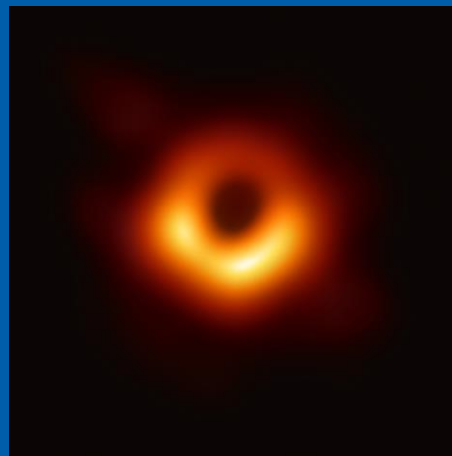
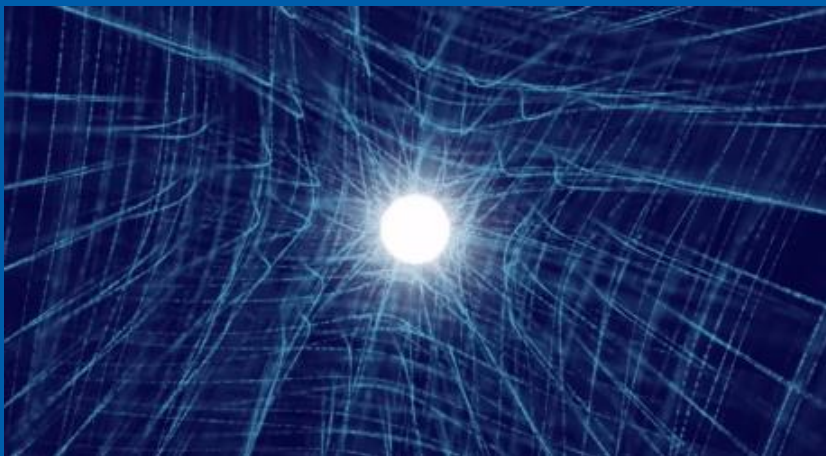
- 10/27/1930 - 01/17/2026
- Mathematician & astronomer
- Processed NASA satellite data to model the geoid (the exact shape of the Earth)
- Vital contribution to GPS technology



Course Overview

What we will cover:

- Earth, planets, solar systems, galaxies and other cosmic structures
- Motion and gravity
- Light, spectroscopy, stars and all the stuff we can see
- Black holes and other things we can't quite see
- Time, space, spacetime and our expanding universe



What topic are you most excited to learn about?

Syllabus Review

Instructor: Matthew Prest

Section: 04 (5763)

Time: Thursdays 12:00pm - 2:50pm

Office: North Building 1220A (<https://us06web.zoom.us/j/3163865896>)

Email: mprest@gradcenter.cuny.edu

Virtual Office Hours: Mondays 4:00pm - 6:30pm (<https://us06web.zoom.us/j/3163865896>)

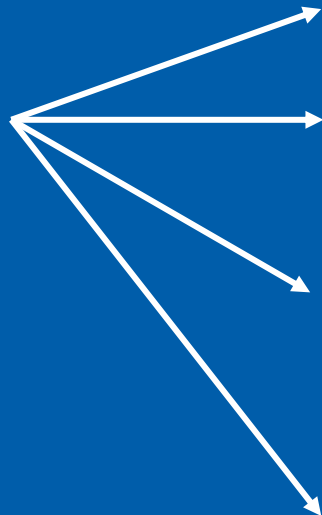
In-person Office Hours: Appointments welcome, send me an email!

Department Office Information: Room 1225 HN, Tel: (212) 772-5248.

In-Person Lab Location: HN-C115

Syllabus Review

1. **Sign up for the Pearson online textbook!** (instructions in accompanying document, and attempt lab 00 for extra credit!)
2. **Print In-person lab documents** (available on Brightspace)
3. **Add the following in-person lab dates to your calendar:**
 - a. **Thursday Feb 19th** (Lab 3, Diameter of the Moon)
 - b. **Thursday March 5th** (Lab 5, Simple Pendulum)
 - c. **Thursday March 26th** (Lab 8, Spectroscopy)
 - d. **Thursday April 23rd** (Lab 10, Lenses)
 - e. **Thursday May 14th** (Make-up Lab Session)



Date:	Lab Details:
Thursday January 29th	Lab 01: Scales of the Universe
Thursday February 5th	Lab 02: Reason for the Seasons
Thursday February 12th	<i>No Lab - College Closed</i>
Thursday Feb 19th	Lab 03: Diameter of the Moon
Thursday February 26th	Lab 04: Motion and Gravity
Thursday March 5th	Lab 05: Simple Pendulum
Thursday March 12th	Lab 06: Orbital Motion
Thursday March 19th	Lab 07: Light & Spectroscopy
Thursday March 26th	Lab 08: Spectroscopy
Thursday April 2nd	<i>No Lab - Spring Break</i>
Thursday April 9th	<i>No Lab - Spring Break</i>
Thursday April 16th	Lab 09: Light & the Doppler Effect
Thursday April 23rd	Lab 10: Lenses
Thursday April 30th	Lab 11: Black Holes
Thursday May 7th	Lab 12: Hubble's Law
Thursday May 14th	Make-up In-person Lab
Thursday May 21st	<i>No Lab - Final Exams</i>

Lab Reports

What makes a good lab report?

Lab Reports

What makes a good lab report?

- Precise instructions (what did you actually do?)
 - Don't worry if you got the "wrong" result!
- Impersonal descriptions
 - It's great that you had fun doing the experiment but we don't need to record that in the lab report! Focus on the details!

Lab Reports

A good lab report is like a good recipe!

- Easy to follow, clear instructions with all of the details you need to recreate the dish exactly as you made it, even if it didn't turn out perfect!
- That's how scientists build trust with one another, by doing experiments that are reproducible!



Lab Reports

Lab reports come last!

- Complete the essay questions once you have finished the tutorial!
 - Remember that science isn't about being "right" it is about making predictions and honestly assessing the results based on evidence.

Pearson Online Platform

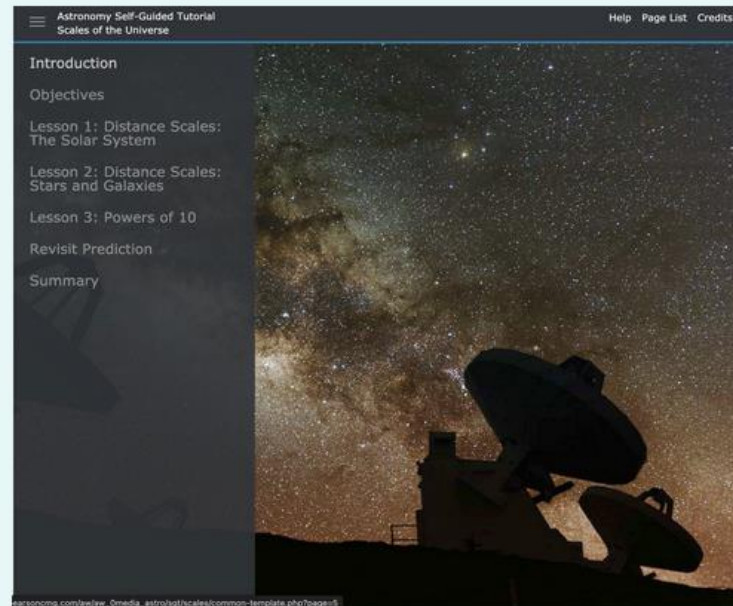
What you should see:

+ Lab 01: Scales of the Universe		
Due: 02/05/26 11:59 AM	15 items, about 110-135 minutes	Partial Credit
Score: 0.0 %		
Items	Possible Points	
1. Scales of the Universe Lab Report	25 pts	
2. Key Concept: Astronomical Distance Units	5 pts	
3. Ranking Task: Size and Distance Scales of Levels of Structure in the Universe	5 pts	
4. Ranking Task: Astronomical Distances and Light-Travel Time	10 pts	
5. Ranking Task: Looking Back in Space and Time	3 pts	
6. Problem 1.42 - Enhanced - with Hints and Feedback	8 pts	
7. Problem 1.50 - Enhanced - with Hints and Feedback	8 pts	
8. Problem 1.52 - Enhanced - with Hints and Feedback	5 pts	
9. Problem 1.41 - Enhanced - with Hints and Feedback	5 pts	
10. Problem 1.51 - Enhanced - with Hints and Feedback	5 pts	
11. Problem 1.53 - Enhanced - with Hints and Feedback	5 pts	
12. Problem 1.45 - Enhanced - with Hints and Feedback	4 pts	
13. Problem 1.48 - Enhanced - with Hints and Feedback	5 pts	
14. Problem 1.15 - Enhanced - with Hints and Feedback	4 pts	
15. Problem 1.39 - Enhanced - with Hints and Feedback	3 pts	

Pearson Online Platform

What you should see:

Launch the [Scales of the Universe](#) tutorial. As you go through the tutorial, answer the ungraded questions in the tutorial and the graded essay questions below.



Pearson Online Platform

Introduction

Objectives

Lesson 1: Distance Scales: The Solar System

Lesson 2: Distance Scales: Stars and Galaxies

Lesson 3: Powers of 10

Revisit Prediction

Summary



Scales of the Universe

What's the difference between a solar system and a galaxy?



Scales of the Universe

What is a lightyear and how big is it?



Reminders

1. Complete the Tutorial first!

- This is best done now in class where I am available to assist

1. Attempt the lab report essay questions last!

- These should be based on what you have covered in the tutorial.

1. Previous week's lab report due at the beginning of the next lab!

- Final grades are based on best 6 online labs plus each of the 4 in-class labs

1. Grades will be posted to Brightspace

- Check now that you have access to our course!

1. Office hours

- Office hours are available on Zoom Mondays 4-6pm (same zoom link as class) as well as by appointment either in-person on or zoom. Come say hi, I'm here to help!